

## NEET Important Chapter – Work, Energy and Power

### Work, Energy and Power: An Important Concept for NEET

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This chapter of work, energy and power deals with the prior revision of all the concepts we studied in our lower grades to build the connection between the advanced parts of the same. In work, energy and power, we have already studied the formulae which describe physical entities like energy, force and work creating the basics of other derived quantities.

In Physics, work is defined as the scalar product of force and displacement which sometimes is also considered as the work done by a constant force.

$$W =$$

$$\text{Or } W = Fs$$

Here  $F$  is the force and  $s$  is the displacement whereas  $\theta$  is the angle acting between both of them. This relation helps to understand the various related terms like the angle between force and displacement, and what is the nature of work associated with these two quantities.

Moving further, the chapter brings into light the concepts of work done by variable force, the Work-Energy theorem and some other forms of energy. We have also studied the conservative and non-conservative forces and the stability of equilibrium when different forces act in the system, Gravitational Potential energy motion in the vertical circle, energy related to the spring system, etc.

Now, let us move on to the important concepts and formulae related to [NEET](#) exams along with a few solved examples.

## Important Topics of Work, Energy and Power

- Work and its unit
- Work done by a Variable force
- Work Energy Theorem
- Conservative Forces
- The potential energy of spring
- Gravitational Potential Energy
- Vertical Circular Motion
- Elastic Collision
- Inelastic Collision
- Energy and its various types
- Conservation of Mechanical Energy
- Energy and its various types

## Important Concepts of Work, Energy and Power

| Name of the Concept | Key Points of the Concept |
|---------------------|---------------------------|
|---------------------|---------------------------|



|  |                                                                                                                                                                |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>by finding the area under the curve.</p> <ul style="list-style-type: none"> <li>• Or we can say area under force displacement graph =</li> <li>•</li> </ul> |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Energy and its various types      | <ul style="list-style-type: none"> <li>• Energy is another name of work done. Contrarily, we can say that it refers to the capability of the body to do some kind of work.</li> <li>• Kinetic Energy - This energy is related to the virtue of motion of a body, having formula as <math>K.E = \frac{1}{2}mv^2</math></li> <li>• Potential Energy - This energy is related to the change in position of the body in some field with the formula <math>= mgh</math>.</li> </ul> |
| Work Energy Theorem               | <ul style="list-style-type: none"> <li>• This is one of the most important theorems stating that the work done by a net force in displacing a body is the same as the change in kinetic energy of the body.</li> <li>• <math>W = \Delta K</math> This formula depicts the entire statement stated above.</li> </ul>                                                                                                                                                            |
| Conservation of Mechanical Energy | <ul style="list-style-type: none"> <li>• This theorem states that the sum of kinetic and potential energy of the body is mechanical energy i.e. <math>E = K + V = \frac{1}{2}mv^2 + mgh</math>.</li> <li>• If the force acting on the body is conservative then we can say that mechanical energy of that body is conserved.</li> </ul>                                                                                                                                        |

## Conservative Forces

- If the work done by a body in displacing it from one position to another is independent of the path followed then that force is called Conservative Force.

- Another major characteristic of this force is work done in this case is always zero.

|                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                   | <ul style="list-style-type: none"> <li>• Examples of conservative forces are gravitational force, spring force and electric force.</li> <li>• If <math>\Delta U = 0</math> then</li> <li>• <math>\Delta U = \text{_____}</math></li> <li>• Hence, a conservative force is a negative gradient of potential energy.</li> </ul>                                                                                                                                                                                    |
| <p>Other Forms of Energy:-<br/>The potential energy of spring</p> | <ul style="list-style-type: none"> <li>• This energy is related to the application of force done on a spring system. The external force on the spring produces a restoring force inside the spring due to compression or extension. <math>F = -kx</math></li> <li>• Hence, the energy stored in spring system is given by relation :</li> <li>• <math>U = \frac{1}{2}kx^2</math></li> </ul> <p>Where <math>k</math> is the spring constant and <math>x</math> is the compression or extension in the spring.</p> |

## Gravitational Potential Energy

- This is another example of conservative force stated as whenever a body is raised to some height, it acquires some kind of energy due to a change in its position. This energy is called Gravitational Potential Energy.
- Since the raising of the body is against the direction of earth's gravity, the force acting here is negative.
- The work done here is written as  $W = mgh$ .
- $U = - \frac{GMm}{r}$  for two bodies having mass  $M$  and  $m$  separated by distance  $r$

## Vertical Circular Motion

- Considering a body of mass ' $m$ ' is tied with a string (massless) and imparts a velocity at the lowest point such that it completes the circular trajectory.
- $A$  is the lowest point,  $V_A$  is the velocity imparted to the particle at the lowest point and the length of the string is ' $R$ '



|  |                                                                                                                                                                                                                                                                                                                |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <ul style="list-style-type: none"> <li>• Such that tension acting at C(highest point)=<br/>whereas tension at lowest point =<br/>.</li> <li>• Minimum velocity <u>required</u> by a body to complete circular motion is</li> <li>• Minimum <u>velocity</u> required by a body to reach at point B =</li> </ul> |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                         |                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Power                   | <ul style="list-style-type: none"> <li>• Power is defined as a dot product of force and velocity or it is also written as the rate of work done.</li> <li>• <math>P = \frac{W}{t}</math> or <math>P =</math></li> <li>• The dimensional formula of power is with the measuring unit as Watt.</li> </ul>                                             |
| Collision and its Types | <ul style="list-style-type: none"> <li>• Collision is defined as the sticking of two bodies after colliding with each other or if the force exerted by one body affects the path of another body then the two bodies are said to be colliding.</li> <li>• Collision is classified under two categories</li> <li>• Elastic and Inelastic.</li> </ul> |



|                     |                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Elastic Collision   | <ul style="list-style-type: none"> <li>• Those collisions in which the laws of total energy and momentum remain conserved along with conservation of kinetic energy are called elastic collisions.</li> <li>• In elastic collision, the relative velocity of separation is equal to relative velocity of approach.</li> <li>• Here, the value of coefficient of restitution is 1 i.e. <math>e=1</math></li> </ul> |
| Inelastic Collision | <ul style="list-style-type: none"> <li>• Here only the laws of total energy and momentum remain conserved but kinetic energy conservation doesn't take place.</li> <li>• In inelastic collision, the value of the coefficient of restitution lies between 0 and 1.</li> </ul>                                                                                                                                     |

### List of Important Formulae

| Sl. No | Name of the Concept           | Formula                                                |
|--------|-------------------------------|--------------------------------------------------------|
| 1.     | Work                          | $W=Fs$                                                 |
| 2.     | Work done by a Variable force |                                                        |
| 3.     | Energy and its various types  | Kinetic Energy (K.E) = —<br>Potential Energy = $mgh$ . |
| 4.     | Conservative Forces           | —                                                      |
| 5.     | Work- Energy Theorem          | — —                                                    |

|    |                                                              |                                                                                                                                       |
|----|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 6. | Conservation of Mechanical Energy                            | $E = K + V = \frac{1}{2}mv^2 + mgh$ .                                                                                                 |
| 7. | Other forms of Energy<br>-<br>The potential energy of spring | Work done in spring system<br>$= -\frac{1}{2}kx^2$                                                                                    |
| 8. | Gravitational Potential Energy                               | The Work done here is written as $W = mgh$ .<br>$U = -\frac{GMm}{r}$ for two bodies having mass $M$ and $m$ separated by distance $r$ |

|    |                          |  |
|----|--------------------------|--|
| 9. | Vertical Circular Motion |  |
|----|--------------------------|--|

|     |                                       |                                                                                                                                                                                                                                                                                                                  |
|-----|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                                       | <ul style="list-style-type: none"> <li>• Tension acting at C(highest point)=<br/>whereas tension at lowest point<br/>= .</li> <li>• Minimum velocity required by <u>body</u> to<br/>complete circular motion is</li> <li>• Whereas minimum <u>velocity</u> required by body<br/>to reach at point B =</li> </ul> |
| 10. | Power                                 | $P = \frac{W}{t}$ or $P = F \cdot v$                                                                                                                                                                                                                                                                             |
| 11. | Elastic Collision in<br>one dimension | <p>Here, the formula of final velocity after collision is</p> $v_1 = \frac{m_1 - m_2}{m_1 + m_2} u_1 + \frac{2m_2}{m_1 + m_2} u_2$ $v_2 = \frac{2m_1}{m_1 + m_2} u_1 + \frac{m_2 - m_1}{m_1 + m_2} u_2$                                                                                                          |

## Solved Examples from Work, Energy and Power

**1. A porter lifts a suitcase weighing 20kg from the platform and carries it in his hand 2.0m above the platform. Calculate the work done by the porter on the suitcase.**

**Sol:**

Given,

(negative sign is due to motion against gravity)

negative sign is due to motion against gravity,

**Hence**  $\quad \quad \quad = -400 \text{ Joule.}$

**Key Point:** Here, only the work done against gravity will act since the kinetic energy of the suitcase is zero.

**2. An elevator weighing 500kg is to be lifted up at a constant velocity of 0.20m/s. What would be the minimum horsepower of the motor to be used?**

**Sol:**

Given,  $m = 500\text{kg}$ , Velocity =  $0.20/\text{s}$

$g = 10 \text{ m/s}^2$

We have to calculate the power in horsepower.

We know

Where  $F$  is force acting on the elevator. Thus,

Hence,

$$= (500)(0.20)(10)$$

$$= 1000\text{W}$$

And

**Therefore, Power in hp =  $1000/760 = 1.315\text{hp}$**

**Key Point:** Here only power's formula is the main thing that students should remember.

### Previous Year Questions from NEET Paper

**1. A force  $F = 20 + 10y$  acts on a particle in  $y$  direction where  $F$  is in newton and  $y$  in metres. The work done by this force to move the particle from  $y=0$  to  $y=1\text{m}$  is (NEET 2019)**

**Sol:**

Given,  $F = (20 + 10y) \text{ N}$

We have to calculate the work done from  $y=0$  to  $y=1\text{m}$ .

Here, the force acting is variable force; hence

$$W =$$

$$=$$

$$= \quad \text{---}$$

$$= 25 \text{ J}$$

**Hence, the work done by this force to move the particle from  $y=0$  to  $y=1\text{m}$  is 25 J.**

**Trick:** We can solve the question using the formula of power.

**2. The work done to raise a mass  $m$  from the surface of the earth to a height  $h$ , which is equal to the radius of the earth, is: (NEET 2019)**

**Sol:**

The initial potential energy of the mass at the surface of the earth having radius  $R$  is

$$U_i = \text{---}$$

When mass  $m$  is raised to height  $h$ , the final energy becomes

$$U_f = \text{---}$$

Hence, the work done here will be the change in kinetic energy  $= U_f - U_i$

$$= \text{---} - \text{---} \quad (\text{here } h = r)$$

$$= \text{---} - \text{---}$$

$$= \text{---}$$

$$\text{But } g = \text{---}$$

**Hence, the final work done is ---**

**Trick:** We can solve the question with the help of the work-energy theorem.

### Practice Questions

1. A cyclist comes to a skidding stop at 10m. During this process, the force on the cycle due to the road is 200N and is directly opposed to the motion. How much work does the road do on the cycle?

**Ans -2000J**

2. A ball with a velocity of 5 m/s impinges at an angle of  $60^\circ$  with the vertical on a smooth horizontal plane. If the coefficient of restitution is 0.5, find the velocity and direction after the impact.

**Ans:  $v = 0.3\text{m/s}$  and the direction will be perpendicular to the surface**

### Conclusion

In this article, we have provided all important information regarding the chapter Work, Energy and Power. Some of these include important formulae, concepts, etc. Students should work on more solved examples, sample papers and previous year questions for securing good marks in the NEET exam.

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